

Recall univ prop. of a basis $B = \{v_\alpha\}_{\alpha \in I}$ of V .

Given any V -space U and a family $c \subset U$ (indexed by same I)

$\exists!$ linear map $f: V \rightarrow U$ with $f(v_\alpha) = u_\alpha$ $\{u_\alpha\}_{\alpha \in I}$

How? We are forced to define $f(c_1 v_{\alpha_1} + \dots + c_n v_{\alpha_n}) = c_1 u_{\alpha_1} + \dots + c_n u_{\alpha_n}$. Check: this is linear.

Direct sums: external^① and internal^②

① Let $W_1, W_2, \dots, W_n$ be any v-spaces

Their "external direct sum" $= W_1 \times W_2 \times \dots \times W_n$ as a set, made into a vector space by slotwise $+$ and scalar mult. Rmk: $\mathbb{R}^n = \underbrace{\mathbb{R}^1 \times \dots \times \mathbb{R}^1}_{n \text{ times}}$

Observe a universal property of this construction!

Given any vector space U and linear $W_i \xrightarrow{f_i} U$, $\exists!$ linear map

Observe a universal property of this construction! n times
 Given any vector space U and linear $W_i \xrightarrow{f_i} U$, $\exists!$ linear map

$$f: W_1 \times \cdots \times W_n \rightarrow U \quad \text{st.} \quad f(\underbrace{w_1, \dots, w_n}_{\bar{w}}) = f_1(w_1) + \cdots + f_n(w_n)$$

Linearity of f : easy check. $f(\bar{w} + \bar{w}') = f(\bar{w}) + f(\bar{w}')$, $f(c\bar{w}) = c f(\bar{w})$

Note: A basis for $W_1 \times W_2 \times \cdots \times W_n$ can be obtained as

(basis for W_1) $\times 0 \times \cdots \times 0 \cup 0 \times (\text{basis of } W_2) \times 0 \cdots \times 0 \cup \cdots$
 $\dim(W_1 \times \cdots \times W_n) = \sum_{i=1}^n \dim W_i$ We call $W_1 \times \cdots \times W_n$ the direct sum of W_i 's.

Notation $W_1 \oplus W_2 \oplus \cdots \oplus W_n$ $\left\{ \begin{array}{l} (w_\alpha)_{\alpha \in I} \in \bigoplus W_\alpha \\ \text{exactly when} \end{array} \right.$

Rmk. For family $\{W_\alpha\}_{\alpha \in I}$ with infinite index set I , $\bigoplus_{\alpha \in I} W_\alpha \subsetneq \prod_{\alpha \in I} W_\alpha$ only finitely many $W_\alpha \neq 0$

② Internal direct sum Suppose $\{W_\alpha\}_{\alpha \in I}$ is a family of subspaces of a v-space W .

Defⁿ ① $\text{Span } \{W_\alpha\}_{\alpha \in I} = \text{Span } \bigcup_{\alpha \in I} W_\alpha \stackrel{\text{note}}{=} \left\{ W_1 + \dots + W_n \mid \begin{array}{l} W_i \in W_{\alpha_i} \\ \text{for some } \alpha_i \in I \end{array} \right\} \stackrel{\text{notation}}{=} \sum_{\alpha \in I} W_\alpha$

② $\{W_\alpha\}_{\alpha \in I}$ is called independent if $\left. \begin{array}{l} W_1 + \dots + W_n = 0 \\ \text{with } W_i \in W_{\alpha_i} \end{array} \right\} \Rightarrow \text{each } W_i = 0$.

Earlier defⁿ of independence is just a special case of this where each W_α is 1-dim^l

③ Combine ① & ② to define 'internal direct sum' (Tomorrow)